

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 1786:2008

2008

HIGH STRENGTH DEFORMED STEEL BARS

SPECIFICATION

ICS 77.140.15

BIS 2008

Indian Standard

1 SCOPE

This standard covers hot rolled high strength deformed steel bars and wires for concrete reinforcement. These bars have deformations on the surface for better bonding with concrete.

1 GRADE DESIGNATIONS

Grade	YS (Min)	UTS (Min)	Elongation (Min)
Fe 415	415 MPa	485 MPa	14.5%
Fe 500	500 MPa	545 MPa	12%
Fe 550	550 MPa	585 MPa	10%
Fe 600	600 MPa	635 MPa	8%

2 CHEMICAL COMPOSITION

Element	Fe 415	Fe 500	Fe 550	Fe 600
Carbon (max)	0.30%	0.30%	0.30%	0.30%
Phosphorus (max)	0.060%	0.055%	0.050%	0.045%
Sulfur (max)	0.060%	0.055%	0.050%	0.045%
Carbon Equivalent (max)	0.42%	0.44%	0.46%	0.48%

3 TOLERANCES

- Weight tolerance: $\pm 5\%$ of nominal weight
- Diameter tolerance: $\pm 0.7\%$ for bars up to 20mm
- Diameter tolerance: $\pm 5\%$ for bars over 20mm
- Length: standard lengths of 6m, 9m, 12m (± 50 mm)

4 TEST METHODS

- Tensile Test: IS 1608 (Part 1) - Room Temperature Test
- Bend Test: IS 1599 - Guide for Bend Test
- Re-bend Test: IS 1599
- Chemical Analysis: IS 228 (Part 1) - Wet Chemical Method

5 MARKING

Each bar shall be legibly and durably marked at intervals not exceeding 150 mm with: (a) Grade designation (Fe 500, etc.), (b) Name/abbreviation of manufacturer, (c) IS 1786 designation, (d) Size of bar.